Supervisor Binary Interface Extension proposal

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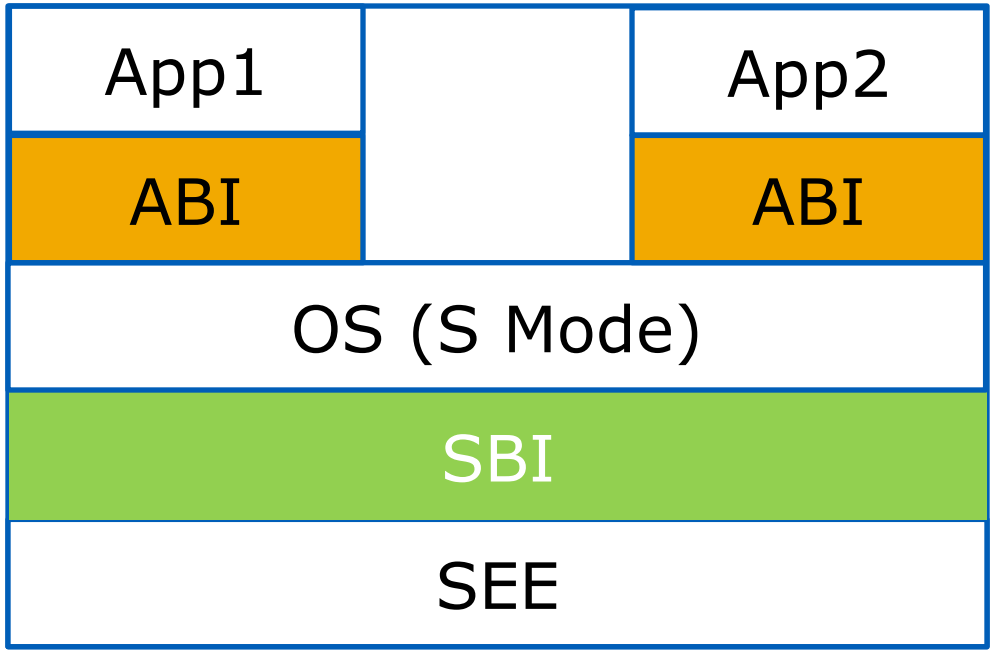
Supervisor Binary Interface (SBI)

- Clear separation between Supervisor & Machine Mode
- Helps to run single OS image across different SEE
- Currently provided by Berkeley Boot Loader (BBL)
- Calling convention
  - S mode traps into M mode using **ECALL**
  - Arguments via a0-a2
  - SBI call type via a7
  - a0 is clobbered register
- Documentation available at
  - https://github.com/riscv/riscv-sbi-doc



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Current status

- Fixed
- No backward compatibility
- Not standardized
- No standard return/error value

Type	Function	Function ID
Timer	sbi_set_timer	0
Console	sbi_console_putchar sbi_console_getchar	1 2
IPI	sbi_clear_ipi sbi_send_ipi	3 4
Memory Model	sbi_remote_fence_i sbi_remote_sfence_vma sbi_remote_sfence_vma_asid	5 6 7
Shutdown	sbi_shutdown	8

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SBI Extension Proposal - Objective

- Make it extendable & backward compatible
- Shouldn’t be a kitchen sink for all hardware abstractions
- Continue to remain minimalistic
- Make it modular to facilitate easy discussion/debate
  - Focus on Legacy (v0.1) & Base SBI
  - Power management/Vendor extension to follow
- Define standard error codes
  - A mapping function between SBI error code & OS specific error
- Ongoing discussion
  - V1
    - http://lists.infradead.org/pipermail/linux-riscv/2018-October/001974.html
  - V2
    - http://lists.infradead.org/pipermail/linux-riscv/2018-November/002171.html

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SBI extension proposal – Calling convention

- Function ID numbering scheme
  - Static binding
    - Bit[31:24] = Function Type Set
    - Number Bit[23:0] = Function Number within Function type
  - Dynamic binding
    - DT Based dynamic probing individual functions

Function Type	Function ID	Description
Base Function	0x0	Base APIs mandatory for any SBI version
HART PM APIs	0x1	Hart UP/Down/Suspend APIs for per-Hart power management
System PM APIs	0x2	System Shutdown/Reboot/Suspend for system-level power management
Vendor APIs	0xff	Vendor specific APIs

- Return Values
  - Just single return value with value (>=0) & error (<0)
  - Return a struct with a value & error (ABI says both a0 & a1 are return values)
  - Error strings in per hart local area?

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SBI extension proposal - Scope

SBI Extension	Function	Type
Base Function	sbi_get_version sbi_check_api or get_feature()	Mandatory
HART PM APIs	sbi_hart_up sbi_hart_suspend sbi_hart_down sbi_hart_get_state	Optional
System PM APIs	sbi_system_shutdown sbi_system_reset	Optional
Vendor APIs	TODO	Optional

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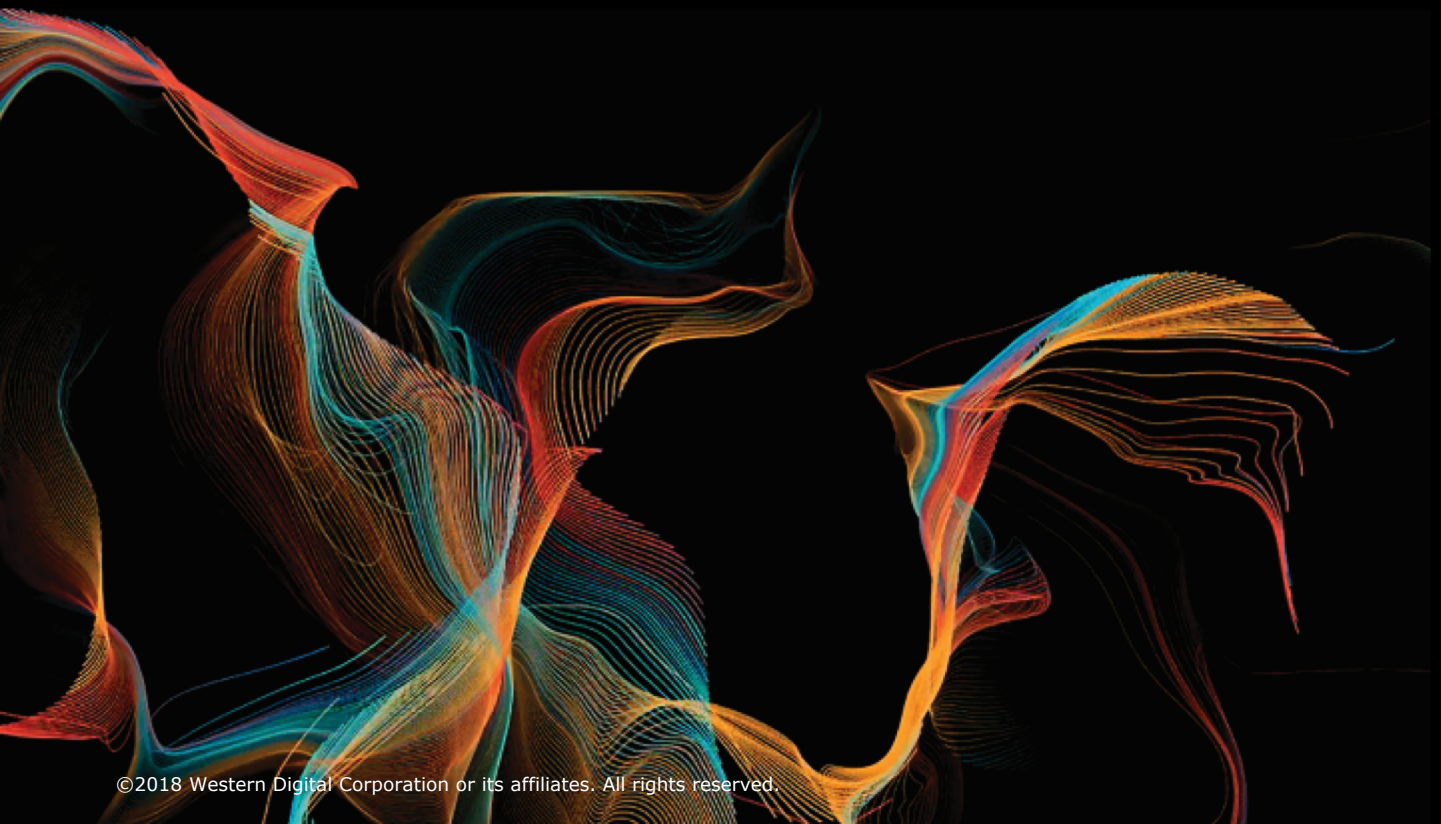

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OpenSBI project

- BBL/Coreboot provides separate SBI implementation
- More fragmentations going forward considering vendor specific usage
- Difficult to maintain & track the SBI changes
- OpenSBI to rescue!!
  - Create a open separate SBI implementation project
  - SBI implementation built as static library that any boot loader(both S mode & M mode) can use
  - Provides a reference implementation that potentially can replace BBL as well
  - This reference implementation can produce firmware binary that can be used as in M mode directly
- Recommended License ?
- Any feedback ?

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Feedbacks

- u32 sbi_check_api(unsigned long start_api_id, unsigned long count)
  vs
  u32 sbi_get_version(void)

- Treat SBI function return & global error separately
  - No global state should be maintained
  - pass by reference
  - return a struct containing err & return value (My choice!!)
  - Should we change all the mandatory existing functions as well to reflect that ?

- May be rename it to a generic interface instead of Supervisor Binary Interface
- Anything else ?

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Etrap call

- lui t0,#ecallTableBase #or auipc
- slli t1,a6,2 #3 for 64 bit
- add t0,t0,t1
- jr,nnn(t0) #lo bits of ecallTableBase, if not 4k aligned
- # called function does the mret

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